Electrodynamic tether in low Earth orbit

Pushing and pulling against a planetary magnetic field, with energy and momentum ledgers — CLAUDE.md §3f

kronspice simulation report · 2026-11-07 · milestone M9n; declared v, B and contactor values

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Summary

A vertical aluminium wire moves east at orbital speed through the northward geomagnetic field. The Lorentz force on its charges, computed by grade selection (*em.MotionalEMF*), gives an EMF (v×B)·L along it, and the circuit closes through the ionosphere by plasma contactors at the two ends. The same loop runs as a generator (the orbit pays for the load) or as a thruster (a supply drives current against the EMF). The force's reaction is booked in the momentum ledger, as every push–pull design must (CLAUDE.md §3f). NASA flew this circuit three times (PMG 1993, TSS-1 1992, TSS-1R 1996) and designed it twice more (ProSEDS, the ISS reboost study); their numbers are set beside the example's.

- **EMF** from the Lorentz force: 1155.0 V, equal to vBL.
- **Generator** (500 Ω load): 1.330 A, 884 W to the load, drag 0.199 N; F·v = −1536 W, so the orbit pays.
- **Thruster** (3000 V supply): −5.004 A, 15,012 W from the supply, thrust 0.751 N, 5780 W into the orbit: 0.05 N/kW.
- **Ledgers:** every energy and momentum identity holds to 10^{-9}; the energy ledger closes to 10^{-12}; the reaction force lands on the field and ionosphere.
- **Against NASA:** the example's 231 V/km is an upper end (v perpendicular to B at 30 μT; TSS-1R measured 177 V/km), and its 0.05 N/kW is half the ISS study's 0.1 N/kW, because 9.2 kW of the 15 kW is I²R heat in the linear contactor stand-ins.

The construction

Declared, not looked up:

input	value
orbital speed v	7.7 km/s
magnetic field B	30 μT (order of magnitude in LEO; a real orbit takes B from a cited field model such as IGRF)
tether	5 km of 1 mm aluminium wire (169 Ω)
contactors	100 Ω each, a linear stand-in for their nonlinear current–voltage characteristics

Two modes of the same loop, each solved as a kronspice operating point with the ledger on:

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EMF - tether resistance - contactors - load          (generator)
EMF - tether resistance - contactors - supply > EMF  (thruster)
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The current feels the force ∫I·dl×B. As a generator the force is drag: the power the load receives comes from the orbit's kinetic energy, F·v = −EMF·I. With a supply driving the current against the EMF the force turns into thrust. *em.MomentumLedger* books the force on the tether with its reaction on the geomagnetic field and ionosphere, the exterior source.

This is the circuit NASA's *Tethers in Space Handbook* writes for the electrodynamic tether (Cosmo & Lorenzini 1997, §4): EMF $V = \int(v\times B)\cdot dl$, which for a straight tether perpendicular to the field is $V = L\cdot v\cdot B\cdot \sin\theta$; force $F = \int I\cdot dl\times B$; and loop equation $V_{ind} = I\cdot R + \Delta V_{low} + \Delta V_{up} + \Delta V_{ion} + \Delta V_{load}$, where the end terms are the drops across the contactors' space-charge sheaths or plasma clouds and ΔV_{ion} is the ionosphere's drop, of order 1 to 20 Ω. The handbook notes that the sheaths are the largest impedances in the loop; the two 100 Ω resistors stand in for them, with the ionospheric drop folded in.

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Results

	generator, 500 Ω load	thruster, 3000 V supply
current up the tether	1.330 A	−5.004 A
electrical power	884 W to the load	15,012 W from the supply
force along x	−0.199 N (drag)	+0.751 N (thrust)
F·v	−1536 W: the orbit pays	+5780 W into the orbit
thrust per power	—	0.05 N/kW
momentum ledger	reaction +0.199 N on the field and ionosphere	reaction −0.751 N on the field and ionosphere

Checks in the test (all to 10^{-9}): F·v = −EMF·I in both modes; generator load power plus I²R losses equals EMF·I; thruster supply power equals losses plus F·v; the momentum ledger's net force equals the force on the tether; the energy ledger closes to 10^{-12}.

The export sheet for the generator (*export/tether_generator_results.md*) gives the MNA node voltages −132.957 V (bot), −357.26 V (m), 797.74 V (top), 664.783 V (x) and 0 (y). Automatic tearing finds no cut of this single loop whose pieces each keep a reference (Aitchison 1983, Lemma 2), so only the MNA column is filled.

Against NASA's flights and studies

	this example	PMG (1993)	TSS-1R (1996)	ProSEDS (designed)	ISS reboost study (1998)
tether	5 km, 1 mm Al, 169 Ω	500 m, #18 AWG Teflon-insulated Cu	19.7 km deployed of 20.7 km; 10 strands of 34 AWG Cu on a Nomex core, 0.305 mm FEP insulation, 2.54 mm overall	5 km of 7-strand 28 AWG 1350-O Al, mostly bare, plus 10 km Dyneema	10 km, bare lower section, < 200 kg
orbit	v = 7.7 km/s, B = 30 μT, v perpendicular to B	193 × 869 km, 25.7°	297 km at the break, near the equator	400 km circular, 36°	ISS orbit
EMF	1155 V (231 V/km)	bias ±65 to −130 V applied in series with the I×B EMF	3482 V (177 V/km)	—	about 100 V/km
current	1.33 A generator, 5.0 A thruster	100–300 mA by day, 10–50 mA at night; 0.3 A at 130 V bias	0.97 A just before the break; 1.1 A for 75 s after it	—	—
force	0.199 N drag, 0.751 N thrust	—	—	about 0.4 N drag; 14 km/day descent	0.5 N from 5 kW, 0.8 N from 10 kW
power	884 W load; 15 kW supply	—	—	about 100 W to the spacecraft	5–10 kW from the arrays
ends	two 100 Ω linear contactors	hollow-cathode xenon plasma contactors at both ends, current fully reversible	1.6 m conducting sphere as anode, electron guns and the Orbiter ground as cathode	bare wire collects, hollow cathode emits	bare wire collects, plasma contactor on the ISS emits

What the comparison says about the example's declared inputs:

- **EMF per kilometre.** The example's 231 V/km is the upper end: it takes v perpendicular to B at 30 μT. TSS-1R measured 177 V/km near the equator at 297 km, and the ISS study quotes about 100 V/km for the station's 51.6° orbit. The handbook's sin θ is the difference, which is why a real orbit must take B from a field model.
- **Thrust per watt.** The example's thruster gives 0.05 N/kW against the study's 0.1 N/kW at 5 kW and 0.08 N/kW at 10 kW. Of the 15 kW drawn, 9.2 kW is I²R heat in the 369 Ω of tether and contactor resistance and 5.8 kW goes into the orbit. NASA's design cuts both: a bare tether collects electrons along its length with no end sheath to cross (Sanmartín, Martínez-Sánchez & Ahedo 1993), and a hollow cathode emits them at a drop of tens of volts.
- **The contactors** are the weakest part of the model. TSS-1R's sphere collected currents up to three times what the Parker–Murphy model predicted, at lower satellite potentials than predicted, and Stone & Raitt conclude that

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the system's motion through the plasma affects collection. A nonlinear contactor characteristic fitted to these flight curves is the next step for this example.

- **Reversibility.** PMG demonstrated in flight what the two columns of the results table assume: the same tether and contactors run as a generator and as a motor.

The TSS-1R break

On 25 February 1996 (057/01:29:16 GMT), during STS-75, with 19.7 km of the planned 20.7 km deployed and the EMF at 3482 V, the tether parted at the Orbiter end. The failure board (Szalai 1996) traced it to an arc through a flaw in the insulation that burned the tether through. Gilchrist et al. (1998) report that the current, which had reached 0.97 A, rose to 1.1 A as the bare broken end entered the plasma, and stayed there for about 75 s: a few short strands of copper emitted more current than the Orbiter's conducting surfaces, by a gas-enhanced discharge later reproduced in a plasma chamber. The lesson for this example is the one §3e applies to coils: every design is checked against insulation breakdown at its peak voltage, here the full 1155 V between the tether and the surrounding plasma when the far end is open.

Limits of the model

- v and B are declared (v perpendicular to B at 30 μT), not taken from an orbit and a field model such as IGRF.
- The contactors and the ionosphere are two linear 100 Ω resistors; real sheaths are nonlinear and depend on the motion through the plasma.
- Each mode is a single kronspice DC operating point of one loop at fixed v and B.

Reproducing the results

```
go run ./examples/tether
go test ./examples/tether
KRONSPICE_EXPORT=1 go test ./examples/tether -run TestExport    # netlists, KiCad schematics, results sheets in export/
```

Sources: M. L. Cosmo, E. C. Lorenzini (eds.), *Tethers in Space Handbook*, 3rd ed., 1997, NTRS 19980018321; L. Johnson, M. Herrmann, NASA/TM-1998-208538, NTRS 19980223479; N. H. Stone, W. J. Raitt, Adv. Space Res. (1998), NTRS 19990084046; B. E. Gilchrist et al., Geophys. Res. Lett. 25(4), 437 (1998); K. Szalai, TSS-1R Mission Failure Investigation Board, 1996; L. Johnson et al., AIAA 1998, NTRS 19990107381; J. Ballance, L. Johnson, NTRS 20010020385; L. Curtis et al., STAIF 2002, NTRS 20020048608; I. Katz et al., J. Geophys. Res. 100 (1995); J. R. Sanmartín et al., J. Propulsion and Power 9(3), 353 (1993); L. W. Parker, B. L. Murphy, J. Geophys. Res. 72, 1631 (1967); P. Joot, Theorem 3.16 (reference only); P. W. Aitchison, Quart. Appl. Math. 41, 265 (1983); doc/em-push-pull.md; doc/M9n.md.

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